

ENGINEERING PROFESSIONAL SECTORS IN NEPAL

Chapter 3

- **Public Sectors:**
General Job description of Engineers working in the Public Sectors
- **Private Sectors:**
General Job description of Engineers working in the Private Sectors

The Engineering Council Act 2057

- ▣ The Engineering Council Act 2057 came into existence under the Engineering Council Ain, 2055, Section 37.
- ▣ The Act was initiated for making the engineering profession effective and impressive.
- ▣ Major objectives of the Ain:-
 - a. To make engineering profession more effective
 - b. To regulate systematically and scientifically the engineering profession in Nepal
 - c. To register engineers in the Council according to their qualifications.

OBJECTIVES OF NEC

The objective of NEC is to make engineering profession effective by mobilizing it in a more systematic and scientific and also to register the engineers as per their qualifications. Its duties and responsibilities are:

- **To prepare policies, plans and programs for the smooth functioning of the engineering profession and to execute them**
- **To grant permission and approval to carry out engineering education to those engineering colleges and institutions that meet the required norms and standards and to honor their degrees and certificates**

OBJECTIVES OF NEC

- **To set norms and standards for engineering education in Nepal**
- **To monitor and inspect the quality of engineering education provided by the engineering colleges and institutions**
- **To fix the qualification necessary in order to practice engineering profession and to register their name in the Council**
- **To remove their name from the registration of the engineering council if found to violate the code of ethics.**

Section-2 of the Engineering Council Act:

There will be three separate Register Books-one each for the engineers under the following categories.

- ***A - General Registered Engineers.***
- ***B - Professional Engineers.***
- ***C - Non-Nepalese Registered Engineers.***

The council shall provide a certificate to registered engineers.

The registered engineers have to take Oath in a paper prescribed by the council.

Section -3 of the Engineering Council Act -2055 states that the educational institutions which intend to provide Bachelor, Masters or any postgraduate engineering degree will first need recommendation from the Council before approval by the Government of Nepal.

Section 4, Engineering Council Act, 2055:- Professional Code of Conduct

- a. ***Honesty and Discipline:*** All engineers must provide service/professional jobs without making engineering professional moral low,
- b. ***Loyalty and Morals:*** All engineering professional practicing engineers must be loyal to their organization, employers or clients.
- c. ***Impartiality:*** While practicing engineering professional knowledge and skill, engineers should be impartial in relation to religion, color, gender or caste or any other.
- d. ***Performing only related professional jobs only:*** The engineers should practice their knowledge and skill in engineering within the scope of their education, profession, knowledge and skill.

Professional Code of Conduct as per NEC

- e. *Never do anything that brings low moral in the Engineering Profession:* **Engineers should not come financial or other influences by any change while performing engineering jobs that lowers engineering moral in the society.**
- f. *Personal Responsibility:* **In engineering professions, engineers are responsible for their doings.**
- g. *Name, Rank and Register no. :* **The engineers must produce their name, rank and registration number after their professional work like writing, drawing, design, specification, estimate, etc.**
- h. *Do not publish statements to leave unnecessary influence:* **The engineers shall not publish any notice or writing that brings influence the public and other client unnecessary.**

Section 5 , Engineering Council Act,2055

- ▣ An examining committee comprising of three members formed under the coordination of a Council member shall examine any complaint registered against any engineer in the Council.

System of Provision for private practice and for employee engineers.

Nepal Engineering Council Act,2055 states that :-

No engineer should practice engineering profession without being registered in the Council even if graduated from the institution recognized by the Council. This will be effective after a year from the implementation of the Nepal Engineering Council Act,2055.

Removal of Name from Register Book

The name of Engineers shall not be removed from Register Book except in the following cases:

- 1. Mentally ill**
- 2. Bankrupt in case being unable to pay loans back**
- 3. Violated specified professional conducts and reported to NEC to remove name from Register Book and if decided by two third majority**
- 4.If the court proved the engineer guilty in public case on moral ground**
- 5.By mistake or mischief if anybody without basic qualification has been registered**

Cancellation of the Certificate

- ▣ If the Council decides to remove the name of the engineer from the Register Book , the Registrar will inform the engineer by removing the name from the Register Book and cancelling his/her certificate.

Re-registering the name:

An engineer can apply for re-registration of name in the Register Book after a year his name has been cancelled using prescribed format . But there has to be reasonable causes for re-registration.

If the Council decides to re-register the name , it can register the name of the applicant again.

The Registrar has to provide a certificate of re-registration to the applicant.

NEPALESE LEGAL SYSTEM AS IT AFFECTS THE PRACTICE OF ENGINEERING

- 1. Financial Administrative Directive 2048 (FAR)**
- 2. Public Works Directives (PWA) developed under FAR**
- 3. Related Manuals, Guidelines, Standards and Specifications adopted by Departmental Decisions**
- 4. Contract Act 2056**
- 5. Arbitration Act 2055**
- 6. Contract Management Act 2020**
- 7. Contractor Development Act 2055**
- 8. Procurement Guidelines for Donors developed under FAR**

Engineering Professional Practice Sectors in Nepal

➤ **Public Sectors, Government Organizations like Ministries, Departments, Regional & District Offices, Corporations, Institute of Engineering etc.**

➤ **Private Sector**

Public sectors (ministries) where qualified engineers can practice

- 1. Finance Ministry**
- 2. Industry, commerce and supply ministry**
- 3. Law, justice and parliamentary ministry**
- 4. Agriculture and cooperative ministry**
- 5. Home ministry**
- 6. Population and environment ministry**
- 7. Water resource ministry**
- 8. Physical planning and works ministry**
- 9. Culture, tourism and civil aviation ministry**
- 10. Foreign ministry**
- 11. Land-reform and land management ministry**
- 12. Women, children and social welfare ministry**
- 13. Defense ministry**
- 14. Forests and soil conservation ministry**
- 15. Science and technology ministry**
- 16. Education and sports ministry**
- 17. General administrative ministry**
- 18. Information and communication ministry**
- 19. Local development ministry**
- 20 health Ministry**
- 21. Labor & Transport Ministry**

Corporations and companies which are considered semi-governmental organizations in which apex members are nominated by government and other members through public service commission.

- 1. The National Construction Company Nepal(NCCN)**
- 2. The Nepal Telecommunication Authority(NTA)**
- 3. Nepal Airlines Corporation**
- 4. Nepal Dairy Development Corporation**
- 5. Nepal Industrial Development corporation**
- 6. Nepal Electricity Authority**
- 7. Nepal Rastra Bank**
- 8. Agricultural Development Bank**
- 9. Rastriya Banijya Bank**
- 10. Tribhuvan University**
- 11. University Grant Commission**
- 12. Hetauda Cement Corporation**

Engineering Practice in Government Sector

The duties of engineer are mainly:

- **Design and estimate of specified project, infrastructure etc.**
- **Preparation of technical specification, tender documents, contract document, agreement paper**
- **Evaluate and supervise the project assigned him/her.**
- **Allocation of resource**
- **Working as a member of investigation committee**
- **Planning**
- **Provide suggestion, recommendation as their expertise.**

General Job Description of Fresh Graduates (Gazetted third class) working in Public Sectors

- **To perform preliminary and detail survey, design and estimate**
- **To execute and assign for execution of project works**
- **To conduct various programs for increasing people's capacity.**
- **To report writing of –**
Progress Report , Feasibility Report , Final Report, Monitoring and Evaluation report etc.

General Job Description of Engineers working in Public Sectors (Gazetted Third Class)

- **To execute other jobs planned specifically for engineers as the nature and case be.**
- **To monitor and evaluate ongoing projects**
- **To facilitate donor agencies is involved**
- **To monitor and coordinate the operation and maintenance of facilities.**
- **To execute and perform works and jobs assigned by immediate superiors**

General Job Description of Gazetted Second Class Engineers working in Public Sectors

- ▣ Planning , programming and execution of works
- ▣ Research on technology, cases, various skills for upgrading
- ▣ Administrative activities
- ▣ Financial administrative activities

Private Sector Practices in Nepal

- ▣ In the open market system, there are thousands of private organisations and firms working in engineering sectors
- ▣ Entrepreneurs have used engineers to produce in large quantity in economic investments,
- ▣ Private sector working more efficiently under strict supervision and motivations.
- ▣ More than 30 private engineering colleges affiliated to about 5 universities- governmental and non-governmental.
- ▣ Numerous consultancies, construction companies and firms
- ▣ Many Computer institutes, E-business organisations etc have employed engineers

Engineers working in the Private Sectors

- **Construction Companies – *Class A, B, C, D***
- **Consulting Firms or Consultants**
- **Clients or Employers**
- **Private Engineering Colleges**
- **Computer Institutes**
- **Hospitals**
- **Research Centers**
- **Training Centers**

General Job description of Engineers working in the Private Sectors

- **To coordinate works between stakeholders-clients, consulting and contractors**
- **To layout works, to survey and to estimate**
- **To supervise, monitor, and control works.**
- **To control quality, to assess and report to concerning authorities**
- **To prepare bills as a quantity surveyor**

General Job description of Engineers working in the Private Sectors

- **To plan project and report progress**
- **To prepare technical report and prepare claims if any**
- **To conduct necessary training regarding site work and office organizations system to new staffs**
- **To overall manage of construction project etc.**